

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 31

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Gasification: coal (including deposits too deep to mine) is burnt underground under controlled conditions to produce fuel gases (hydrogen, carbon monoxide, methane) (1) • Allows deep coal to be exploited and gas can be piped / used in gas turbines or for hydrogen with CCS (1) • Liquefaction: coal is converted to liquid hydrocarbons directly with solvents or via gasification and chemical conversion (1) • Provides liquid fuels for vehicles / aircraft that solid coal cannot supply (1)	4
01.2	<i>1 mark for each point.</i> • A solid, ice-like crystalline structure in which methane is trapped within water ice (1) • Found at low temperatures (polar regions / permafrost) or high pressure in ocean sediments around continental margins (1)	2
01.3	<i>1 mark for each point.</i> • Water heating: hot water pumped into sediments melts the hydrate releasing methane (1) • Depressurisation: drilling reduces pressure so methane dissociates from the crystals (1) • CO₂ injection: CO₂ bonds more strongly to the ice and displaces methane (could double as CCS) (1) • Risks: methane not collected escapes to the atmosphere - powerful greenhouse gas / positive feedback; seabed ecology poorly understood; sediment instability / landslides (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187, 190; AQA spec 3.3.4.1.	
02.1	<i>1 mark for the correct answer.</i> • $(120 + 95) - (2 + 45) = 168 \text{ t ha}^{-1}$ (1)	1

Question	Marking guidance	Marks
02.2	<i>1 mark for each point.</i> • Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO₂ (1) • Harvest removes biomass so less dead organic matter is returned; pasture roots and litter constantly add organic matter (1)	2
02.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Low / zero tillage (1) • Adding manure, compost or mulch (1) • Growing cover crops / leaving crop residues (1) • Converting to long-term grass or tree crops in rotation (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	
03.1	<i>1 mark for each point.</i> • Deaths rise as smoke concentration rises, peaking at about 930 per day on 8 Dec when smoke was highest (4.5 mg m⁻³) (1) • Deaths fall as smoke falls but remain above the starting level for several days (time lag / delayed effect) (1)	2
03.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • (930 - 250) / 250 x 100 (1) • = 272% (1)	2
03.3	<i>1 mark for each point.</i> • Anticyclonic conditions: clear skies allowed rapid radiation of heat from the ground at night so the ground and air near it cooled (1) • Low wind velocity so air layers did not mix; London lies in a valley / basin where cold dense air collected: a temperature inversion formed (1) • Smoke from coal fires was trapped below the inversion; moist air cooled to its dew point forming fog; smoke + fog = smog; the smog's high albedo prolonged the inversion (1)	3
03.4	<i>1 mark for each point.</i> • Smoke particles (PM₁₀) are inhaled deeply and irritate the lungs / kill cilia so particles and bacteria are not cleared, causing bronchitis / infections (1) • These groups have weaker lungs / immune systems / less reserve so the extra stress (with sulfur dioxide acting synergistically) was fatal (1)	2
03.5	<i>1 mark.</i> • Clean Air Act (1956) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	
04.1	<i>1 mark for each point.</i> • Spoil contains residual sulfide ores (eg iron pyrites, zinc sulfide) (1) • These react with oxygen in the air spaces between the waste particles (oxidation) (1) • Producing metal oxides and oxides of sulfur which dissolve in drainage water to form sulfuric acid (1)	3

Question	Marking guidance	Marks
04.2	<i>1 mark for each point.</i> - The low pH (3.4) directly damages exposed cells / denatures proteins and dissolves calcium-based exoskeletons (1) - Low pH increases the solubility of zinc (0.02 to 6.5 mg dm^{-3}) which is toxic (inhibits enzymes) (1) - Only a few tolerant taxa survive; the sensitive mayfly larvae (indicator of clean water) are eliminated; dissolved oxygen is not the limiting factor here (1)	3
04.3	<i>1 mark for each point.</i> - Dilution by tributaries / more water (1) - Neutralisation by alkaline rocks (limestone) in the stream bed; zinc precipitates / adsorbs onto sediments as pH rises (1)	2
04.4	<i>1 mark for each point.</i> - Collect the drainage water and pass it through crushed limestone / add lime to neutralise the acid (1) - Higher pH also reduces the solubility of toxic metals so they precipitate and can be removed (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9-3.4.3.2.10.	
05.1	<i>1 mark for each point.</i> - A turbofan at the front forces bypass air around the inner cowling / core jet (1) - The bypass air smooths the flow of the turbulent hot exhaust gases, reducing the turbulence that creates noise (1) - The higher the ratio of bypass air to jet exhaust the quieter the engine (Trent 1000 quietest, Olympus noisiest) (1)	3
05.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Chevron (serrated) nozzles mix bypass air with surrounding air more smoothly (1) - Acoustic liners in the engine cowlings absorb noise (1) - Hush kits fitted to older low-bypass engines (1) - Fairings on undercarriage and wing-flap hinges reduce turbulence; welded rather than riveted panels (1) - Lighter composite aircraft need less thrust (1) - Blended wing with engines on top so the body shields the ground (1)	2
05.3	<i>1 mark for each point.</i> - Constant (3°) descent avoids periods of high thrust in a stepped descent and keeps aircraft higher for longer (1) - Night flights (especially take-offs, which are noisier) are restricted / banned when people are sleeping and background noise is low (1) - Noisier aircraft pay higher charges / use up more quota points and fines for exceeding limits, so airlines are encouraged to buy quieter aircraft (1) - More time to slow down after landing so wheel brakes can be used instead of noisy reverse thrusters (1)	4
05.4	<i>1 mark.</i> Engines run at higher power / maximum thrust and the steep climb angle needs full power (although the aircraft climbs quickly out of earshot) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 288-292; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
06.1	1 mark for the correct answer. • $(44 - 8) / 50 = 0.72 \text{ m yr}^{-1}$ (1)	1
06.2	1 mark for each point. • Abstraction (for irrigation) exceeds the rate of natural recharge by infiltration / percolation of precipitation (1) • The dynamic equilibrium between inflow and outflow has been upset so the stored volume declines (1)	2
06.3	1 mark for each point. • Porosity: a high proportion of the rock volume is space that can hold water (chalk, limestone, sandstone) (1) • Permeability: the spaces are interconnected and large enough for water to flow through (clay is porous but not permeable) (1) • Impermeable rock (granite, clay) beneath to stop water escaping, and permeable rock / soil above to allow recharge (1)	3
06.4	Any 4 from the points below - 1 mark each, maximum 4. • Salt water incursion: seawater flows into the aquifer to replace the fresh water, making it unusable for irrigation (osmotic dehydration of crops) (1) • Subsidence: emptied pore spaces no longer support the rock so it compacts, damaging buildings and pipes (eg Mexico City) (1) • Drying up of groundwater-fed rivers, lakes and wetlands as the water table falls below their beds (1) • Ecological impacts: plants needing high water availability die; wetland animals lost; food chain effects (1) • Reduced supplies: abstraction may suddenly have to stop after years of hidden decline (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
07.1	1 mark for each point. • About 8 nights (1) • The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards; extra samples do not change it significantly (1)	2
07.2	1 mark for each point. • Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size (1) • Sampling at different conditions / times would make results not comparable (1) • Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season) (1)	3
07.3	1 mark for each point. • Accuracy: how close a recorded result is to the real value (1) • Precision: the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm) (1)	2
07.4	1 mark for each point. • A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level (1) • Standardised techniques ensure results collected at different times, places or by different people are comparable (1) • Other variables are controlled or monitored so their effect can be assessed; allows the study to be repeated / checked (reliability) (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
08.1	<i>1 mark for each point.</i> - Minerals can only be exploited where they were deposited (no choice of location), unlike housing, roads or ports (1) - Land take is larger than the mine void: buildings, access roads, overburden and spoil heaps, buffer zones; open-cast mines take far more land than deep mines (1)	2
08.2	<i>1 mark for each point.</i> - Dust: water sprays on roads / blasting areas make particles heavier and clump so they settle (1) - Noise: baffle mounds / embankments absorb and deflect; blasting at set, predictable daytime hours; quieter machinery (1) - Turbid water: sedimentation lagoons where water stands until solids settle before discharge (1) - Leachate: collect drainage water and pass through crushed limestone to neutralise acid and immobilise toxic metals (1) - Amenity: landscaping and tree planting to screen the mine; restoration plans agreed as a condition of planning permission (1)	5
08.3	<i>1 mark for each point.</i> - Landscaping to reduce gradients, compaction, drainage to prevent waterlogging, adding soil / nutrients and planting vegetation whose roots bind the surface (1) - At Aberfan a waterlogged coal-mine spoil heap became unstable and slid into the village (1) - Killing 144 people, mainly children in the primary school (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 139-142; AQA spec 3.2.3.6.	
09.1	<i>1 mark for each point.</i> - Radiation absorbed by tissue produces free radicals (ions with unpaired electrons), mostly from water, which are highly reactive (1) - Free radicals damage DNA / chromosomes in the nucleus causing mutations (cancer, inherited effects) - rapidly dividing cells (skin, gut, bone marrow) most affected (1) - Damage to other cell parts can stop the cell functioning or kill it; low damage may be repaired by the intact nucleus (1)	3
09.2	<i>1 mark for each point.</i> - Acute: effects appearing quickly after a large dose over a short time (radiation sickness, damage to bone marrow / immune system / gut, haemorrhage, rapid death) (1) - Chronic: effects appearing slowly, in proportion to the dose received over a long period (cancers, cataracts) (1)	2
09.3	<i>1 mark for each point.</i> - Exposure: absorption of ionising radiation from a source the person is close enough to (1) - Contamination: physically carrying radioactive material on or in the body, so exposure continues wherever the person goes; sealed sources can expose but not contaminate (1)	2

Question	Marking guidance	Marks
09.4	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Sieverts = grays x radiation weighting factor (20 for alpha) (1) • $0.002 \times 20 = 0.04$ Sv (40 mSv) (1)	2
09.5	<i>1 mark.</i> • The activity of a source (number of decays per second) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 301, 303; AQA spec 3.4.3.2.14.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Neonicotinoids are neurotoxins similar to nicotine; they inhibit the neurotransmitter acetylcholine • High toxicity to insects but low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier • Very toxic to bees: sub-lethal doses reduce navigation ability, foraging and immunity to disease; linked to colony decline • Water soluble and moderately persistent: remain in soil, absorbed by roots and translocated (systemic), so present in pollen and nectar • Persistence may allow build-up in aquifers / soil; broken down fastest in sunlight and by soil microbes • Synergism with some fungicides increases toxicity to bees • Exposure greatest when sprayed onto crops or when seed-coating dust is blown by wind • Control: seed coating rather than spraying; spraying when crops are not in flower / at night; still days to reduce drift • Restrictions on use (EU bans on outdoor use on flowering crops); use of alternative pest control (cultural, biological, integrated control) • Evaluation: reduced pesticide use but neonicotinoid alternatives (pyrethroids) also harm pollinators; trade-off with crop yields	9
	Model answer - Level 3 (8 marks)	

Question	Marking guidance	Marks
	Neonicotinoids are neurotoxins related to nicotine that block the neurotransmitter acetylcholine. They are highly toxic to insects but of low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier - the specificity that made them popular. Bees, however, are insects: even sub-lethal doses impair navigation, foraging and immunity, and this has been linked to colony decline. Because the compounds are water-soluble and moderately persistent they remain in the soil, are absorbed by roots and translocated through the plant (systemic action), so they appear in pollen and nectar that bees collect; persistence may also allow build-up in soils and aquifers, although sunlight and soil microbes break them down. Some fungicides act synergistically with them, increasing toxicity to bees.	
	Exposure is greatest when crops are sprayed or when dust from coated seeds is blown by wind, so impacts are reduced by using seed coatings instead of sprays, spraying only when crops are not in flower or at night when bees are not foraging, and applying on still days to limit drift. The EU has banned outdoor use on flowering crops, and integrated control with cultural and biological methods reduces the need for insecticides altogether.	
	These measures reduce bee exposure, but the alternatives farmers turn to, such as pyrethroids, also harm pollinators, and restrictions can reduce yields, so the trade-off between crop protection and pollination services remains.	
	Weaker answer - Level 2 (4 marks)	
	Neonicotinoids are insecticides that kill insects by affecting their nervous system. They are systemic so they get into the whole plant including the pollen, and bees pick them up when they collect pollen and nectar. This harms bees and can cause colonies to die. To reduce the impact, farmers can spray when crops are not flowering, and some neonicotinoids have been banned in the EU. Other methods of pest control could be used instead.	
	What makes the difference	
	• Level 2 links the systemic property to bee exposure correctly but omits the other properties (solubility, persistence, specificity via the blood-brain barrier, synergism) that the question's 'properties' demands.	
	• Level 3 explains the sub-lethal effects (navigation, foraging, immunity) rather than 'harms bees'.	
	• Level 3 includes the trade-off evaluation (alternatives also harm pollinators, yield costs) that the mark scheme rewards.	
10.2	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> • Biological control (introduced predators / pathogens) (1) • Predator habitats (hedgerows, beetle banks) (1) • Crop rotation (1) • Companion / barrier crops (1) • Sterile male technique / pheromone traps (1) • Genetic resistance (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 265-266; AQA spec 3.4.3.2.8.	

Question	Marking guidance	Marks
11.1	To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society? Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • Renewable features: intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal); low energy density; location; low pollution but manufacturing impacts • Technologies and developments for each resource with examples; new technologies improving usability • Storage and grid management: pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving; the role of electricity as a secondary fuel • Economics and politics: development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution; sustainability principles - natural systems run on low-density renewable energy • Judgement: electricity largely yes; heat, aviation, shipping and high-temperature industry harder; transition period with nuclear / CCS Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Stages of investigation: identifying the issue (Halley Station UV data, Dobson unit measurements), hypothesis formation (Rowland-Molina 1974: persistence, UV dissociation, chlorine reactions), data collection (ground, satellite, balloon and aircraft air sampling detecting ClO), analysis of spatial, temporal and altitude variability, proposal of solutions, enactment • Why Antarctica: low stratospheric temperatures, ice crystals, stratospheric clouds, polar vortex, spring sunlight • Effects of UVB: skin cancer, cataracts, DNA damage, plant and marine damage - motivating action • Solutions: Vienna Convention, Montreal Protocol - phase-out and bans, alternative processes (pump sprays), alternative materials (HCFCs, HFCs, hydrocarbons, HFAs), collection and incineration of CFCs, fund for developing countries • Evaluation of success: ozone hole area / depth stabilising, CFC levels falling; limitations - persistence, remaining ODSs, HFCs as greenhouse gases, decades to full recovery • Comparison with other atmospheric problems (climate change, acid rain) - why ozone was easier; role of peer review, international cooperation, availability of substitutes • Conclusion on the extent to which it is a model for other issues Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113; AQA spec 3.2.1.3.	25